

Дана: ~~103. 10~~

~~1. (46)~~

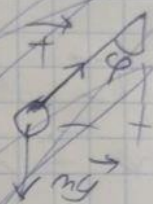
Дано:

$$V = 10^{+1}$$

$$l = 0,5 \text{ м}$$

$\varphi = ?$

Решение:



Дана: ~~10. 10~~

~~1. (5. 6)~~

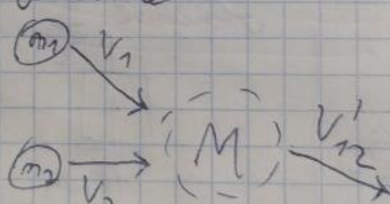
Дано:

$$m_1 = 0,1 \text{ кг}$$

$$m_2 = 0,2 \text{ кг}$$

$$V_1 = V_2 = 0,5 \text{ м/с}$$

$V_{12}' = ?$



По 3. закону сохранения:

$$m_1 V_1 + m_2 V_2 = V_{12}' (m_1 + m_2)$$

$$0,05 + 0,1 = 0,3 V_{12}'; \quad V_{12}' = \frac{0,06}{0,3} = \frac{6}{30} = 0,2 \frac{\text{м}}{\text{с}}$$

Ответ: $0,2 \frac{\text{м}}{\text{с}}$

Дано:

$$m_c = 20$$

$$V_c = 10$$

$$V_n = 2$$

$$m_n$$

$$V_n$$

$$m_c$$

$$10$$

$$10$$

$$V_n$$

$$V_n$$

$$V_n$$

$$V_n$$

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$$V_n$$

Дано:

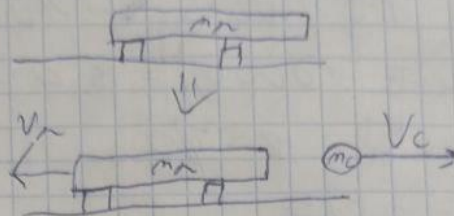
$$m_c = 20 \text{ кг}$$

$$V_c = 102 \frac{\text{м}}{\text{с}}$$

$$V_n = 2 \frac{\text{м}}{\text{с}}$$

m_n

Решение:



По закону сохранения импульса:

$$m_c V_c + m_n V_n = 0$$

$$102 \cdot 20 + 2 \cdot m_n = 0 \quad | :2$$

$$1020 + m_n = 0; \quad m_n = -1020 \text{ кг}$$

Ответ: 1,02 м

Дано:

$$m_n = 1000 \text{ кг}$$

$$V_n = 1 \frac{\text{м}}{\text{с}}$$

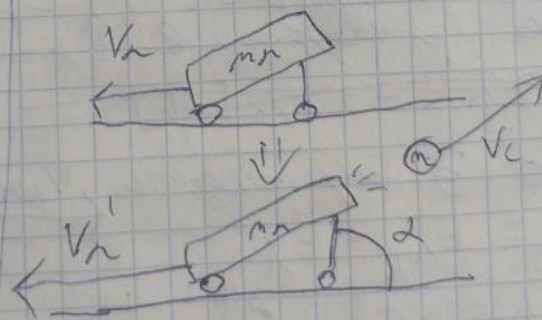
$$\alpha = 60^\circ$$

$$m_c = 10 \text{ кг}$$

$$V_c = 100 \frac{\text{м}}{\text{с}}$$

$$V_n' = ?$$

Решение:



То 3. Corp. unyuzka:

То ош O_x : $P_n = P_n' + P_c$

$$1070 \cdot 1 = 1000 V_n' - 10 \cdot 100 \cdot \cos 60^\circ$$

$$1070 = 1000 V_n' - 500$$

$$1000 V_n' = 1570$$

$$V_n' = 1,57 \frac{m}{c}$$

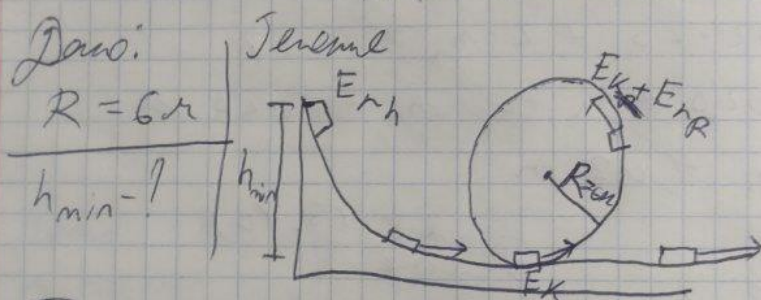
$$\text{Ош: } 1,57 \frac{m}{c}$$

14. (6, 9)

Дано:

$$R = 6m$$

$$h_{min} = ?$$



Меш параметрен $E_n = mgh_{min}$, катге
вакх регелогхн $E_k = \frac{mv_{min}^2}{2}$

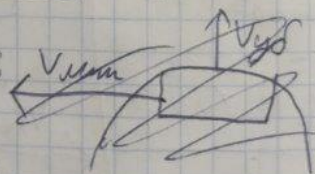
$$\cancel{E_n = E_k} = \cancel{12R} [m]$$

То 3. Corp. Эргун:

$$E_nh = E_nR + E_k$$

$$mgh = 2mgR + \frac{mv^2}{2} \quad | : 2$$

$$2gh = 4gR + v^2$$



То

F_{nR}

mg

v^2

$2h$

Ош: 1

Дано:

$$V_0 = 1$$

$$V' =$$

$$h =$$

$$\frac{100}{2}$$

$$10$$

Ош: 1

№ 3. ~~Механика~~

$$F_{\text{тяг}} = m a_y$$

$$mg = m \frac{v^2}{R} \quad | : m$$

$$v^2 = gR \Rightarrow 2gh = 4gR + gR \quad | : g$$

~~2h~~

$$2h = 4R + R$$

$$h_{\text{min}} = \frac{5R}{2} = \frac{5 \cdot 6}{2} = 15 \text{ м}$$

Ответ: 15 м

Дано:

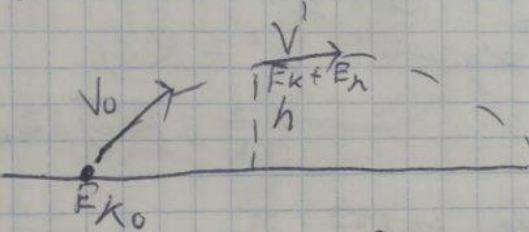
$$V_0 = 10 \frac{\text{м}}{\text{с}}$$

$$V' = \frac{V_0}{2} = 5 \frac{\text{м}}{\text{с}}$$

h = ?

№ 3. (6. 10)

Дано:



№ 3. Corp. frequency:

$$E_{k0} = E_k + E_p$$

$$\frac{mV_0^2}{2} = \frac{mV^2}{2} + mgh \quad | : m$$

$$\frac{100}{2} = \frac{25}{2} + 10^2 h \quad | : 2$$

$$100 = 25 + 20h; h = \frac{25}{20} = \frac{5}{4} = 1,25 \text{ м}$$

Ответ: 1,25 м

nb. (6.11)

Demo:	Jeune:
$L = 2m$	
$V_{min} = ?$	

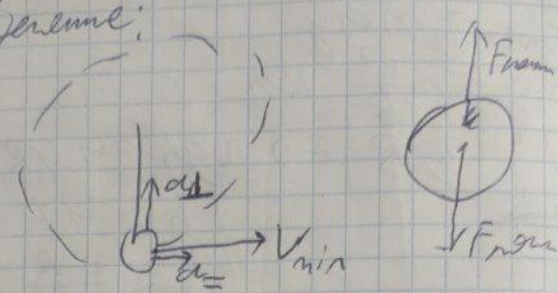


Diagram showing a particle in a circular path. Forces F_{cent} (centrifugal) and F_{grav} (gravitational) are shown. Velocity V_{min} is indicated at the bottom.

То III 3.:

$$F_{cent} = F_{grav}$$

$$mg = m a_{\perp} \quad l.m$$

$$g = \frac{v^2}{R}$$

То IV 3.:

$$V^2 = Rg$$

$$V_{min} = \sqrt{Rg} = \sqrt{2 \cdot 10} = \sqrt{20} \approx 4,5 \frac{m}{c}$$

Answer: $4,5 \frac{m}{c}$

Дана: $l = 0,5 m$

№ 05.10

Jeune: 1. (2.46)

Demo:

$$l = 0,5 m$$

$$V = 10^{-1}$$

$$\varphi = ?$$

То III 3.:

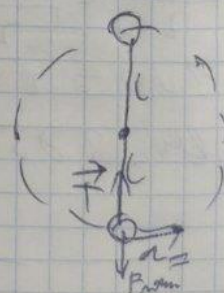
$$T = F_{cent}$$

то 0.1:

$$T \cos \varphi = mg$$

$$\cos \varphi = \frac{mg}{T}$$

~~$$\varphi = \arccos \frac{mg}{T}$$~~



#ХАЛБАЛПАН